

1. What are the most challenging aspects of the coursework task?

The most difficult task in completing this coursework revolved around validating all program inputs without resulting in overly complex code structure. The program needed to execute user choices properly for encryption or decryption operations and validate shift number ranges and file existence verification through multiple operational checks. The conversion from any entered text format into uppercase output proved to be difficult to maintain for user input consistency. The task of file handling required handling of operations to prevent both meaningless errors and accidental destruction of essential file content. The process of maintaining program flow needed structure to satisfy user needs while fulfilling all assignment requirements proved difficult. The task required handling an endless loop of encryption-decryption operations seamlessly to maintain user experience along with complexities that boosted my comprehension of program control systems.

2. How did you go about completing the task?

I completed the work through structured order based on the stated specifications. I established the welcome message to initially present the program to users. The next stage involved developing foundation-level encryption and decryption of single messages with ASCII-based letter shiftings implemented as the core functionality. I added more capabilities to process file content by processing messages line after line while first validating the basic encryption-decryption functions were operating correctly. I performed individual testing of every program section to detect problems before they accumulated. The program includes several loops for validating user inputs where users must enter appropriate data such as mode selections and shift keys and file labels. The program structure included functional breakdowns for better readability and debugging capability. When finishing the coding process I optimized the system to serve new programmers while maintaining best practice features such as easy-to-understand prompts and basic error notification systems.

3. What have you learned over the course of completing this coursework task?

This coursework provided essential understanding about complete input validation as well as proper user interface guidance. The coursework taught me about the essential nature of handling edge situations which include improper data entry together with abrupt user conduct alongside absent file elements. The writing of encryption and decryption functions allowed me to understand better the Python principles behind ASCII code manipulation with modular arithmetic and Python character processing. The experience of working with files taught me to handle text file reading and writing tasks securely. The division of the program content into distinct sections demonstrated to me how modular construction increases program maintainability while enhancing its readability. Loops and error-checking prove essential for creating user-friendly programs because they enhance the entire user experience. My coursework experience confirmed my developing technical abilities alongside better skills to design organized user-focused Python applications that I will need on future projects.